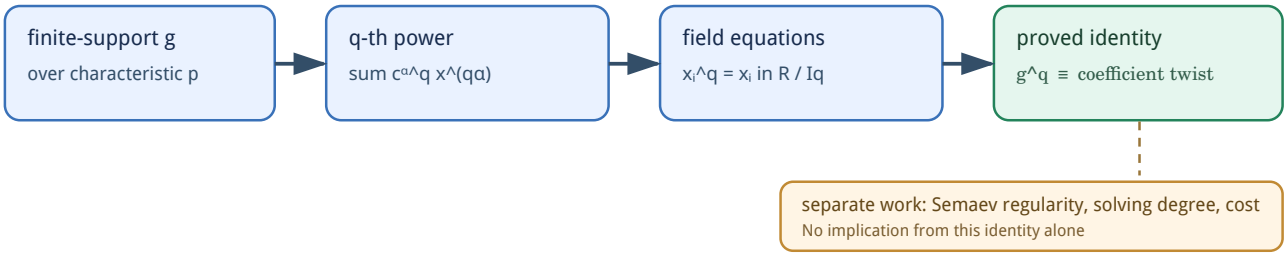
Semaev Frobenius-collapse theorem

Independent semantic review · 2026-11-01 UTC · THM-SEMAEV-FALL1-LEMMA-1

Frobenius collapse: exact proof boundary



Checked result

For characteristic-p coefficient rings and $q = p^e$, the named Lean theorem proves that g^q and the coefficientwise q-Frobenius twist of g agree modulo the ideal generated by $x_i^q - x_i$. The producer's field/prime assumptions are instances of this Lean statement. A blind algebraic derivation and an independent Lean/source inspection agree at this scope.

Evidence

Hosted Lean run	41282682489, source commit 6475884a98b4a0fd0f50ddc7d8b0c82855b74531
Named target	machine_verified; build and axiom audit passed; no forbidden constructs
Axiom audit	propext, Classical.choice, Quot.sound
Source pin	Lean SHA-256 f408877908fc156006c2b24cb38b26610461e03093ff88aadbf11d4bc2c2e398
Review checker	2 reports; all joints owned and attested; blindness and controls passed

Controls and claim boundary

The blind reviewer checked mixed monomials, an extension-field coefficient, and $q = 1$. Characteristic zero fails on $(x+y)^2 - (x^2+y^2) = 2xy$; removing the field equations fails on $x^2 \neq x$ in $F_2[x]$. The Lean theorem does not itself state non-semi-regularity, a solving-degree bound, or a solver-cost result. Corollary 1 is a separate target.

Full report, reviewer attestations, exact file hashes, and reproduction commands: coordination/formal-reviews/semaev-frobenius-collapse-20261101/REVIEW.md. Hosted run: https://github.com/isogeny-labs/crypto-autoresearcher/actions/runs/41282682489